Course Code: BCE-C625
Course Name: Theory of Computation: Classical to Quantum Models

MM: 100 **Time: 3 Hr.** **L T P** **3 1 0**	**Sessional:30** **ESE:70** **Credit :4**

Prerequisites:	Math's Problems on Set Theory, Relations, Operations
Objectives:	• Introduce concepts in automata theory and theory of computation • Identify different formal language classes and their relationships • Design grammars and recognizers for different formal languages • Prove or disprove theorems in automata theory using its properties • Determine the decidability and intractability of computational problems

NOTE:	The question paper shall consist of two sections A and B. Section A contains 10 short type questions of 6 marks each and student shall be required to attempt any five questions. Section B contains 8 long type questions of ten marks each and student shall be required to attempt any four questions. Questions shall be uniformly distributed from the entire syllabus.

Module	Course Content	No. of Hours	POs mapped	PSOs mapped
Module-1	**Regular Languages:** Finite State systems – Basic Definitions – Finite Automation – DFA & NFA – Finite Automaton with e-moves – Regular Expression – Equivalence of NFA and DFA – Equivalence of NFA's with and without e-moves – Equivalence of finite Automaton and regular expressions. Melay and Moore Machine.	**06**	PO1/ PO2	PSO1/ PSO2
Module-2	**Context Free Languages:** Context Free Grammars – Derivations and Languages – Relationship between derivation and derivation trees – ambiguity – simplification of CFG –Derivation trees, sentential forms, right most and left most derivation. Greiback Normal form – Chomsky normal forms – Problems related to CNF and GNF.	**06**	PO1/ PO2/ PO3	PSO1/ PSO2
Module-3	**Pushdown Automata:** Definitions – Moves – Instantaneous descriptions – Deterministic pushdown automata – Pushdown automata and CFL. Designing of PDA. Acceptance by final state, acceptance by null store.	**06**	PO1/ PO2/ PO4	PSO1/ PSO2
Module-4	**Turing Machines:** Turing machines – Computable Languages and functions – Turing Machine constructions – Storage in finite control – multiple tracks – checking of symbols – subroutines – two way infinite tape, Types of Turing machines.	**06**	PO1/ PO2/ PO3/	PSO1/ PSO2
Module-5	**Undecidability:** Chomsky hierarchy of languages, linear bounded automata and context Sensitive language, LR(0) grammar, decidability of problems, Universal Turing Machine, undecidability of posts. Correspondence problem, Turing reducibility, Definition of P and NP problems, NP complete and NP hard problems.	**06**	PO1/ PO2/ PO4/ PO12	PSO1/ PSO2
Module-6	Introduction to Quantum Models of Computation, What is Quantum Computing, Qubits vs Bits: A simple comparison, Basic Concepts: Superposition, entanglement, Quantum Finite Automata (QFA): What changes from DFA/NFA, Quantum Turing Machines: Theoretical extension of classical TM, Quantum vs Classical Models: Comparison of power, limitations,	**10**	PO1/ PO2/ PO4/ PO12	PSO1/ PSO2

	Real-world relevance: Case Study			
Total No. of Hours		**40**		

Learning Outcomes:	• Classify machines by their power to recognize languages. • Employ finite state machines to solve problems in computing. • Explain deterministic and non-deterministic machines. • Comprehend the hierarchy of problems arising in the computer sciences.

Suggested books:

S. No.	Name of Authors /Books /Publisher
1.	"Introduction to Automata Theory Languages and Computation". Hopcroft H.E. and Ullman J. D. Pearson Education
2.	Introduction to Theory of Computation –Sipser 2nd edition Thomson
3.	Introduction to Computer Theory, Daniel I.A. Cohen, John Wiley.
4.	Introduction to languages and the Theory of Computation, John C Martin, TMH
5.	"Elements of Theory of Computation", Lewis H.P. & Papadimitriou C.H. Pearson /PHI.
6.	4 Theory of Computer Science – Automata languages and computation -Mishra and Chandrashekaran, 2nd edition, PHI

CO-PO/PSO MAPPING

Course Outcomes (COs)	Program Outcomes (POs)												Program Specific Outcomes (PSOs)	
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	√	√											√	√
CO2	√	√	√										√	√
CO3	√	√	√									√	√	√
CO4	√	√	√	√								√	√	√